

Developer Architecture & SDLC Documentation

Multilingual AI Business Operating System & Layman Chatbot MVP Architecture

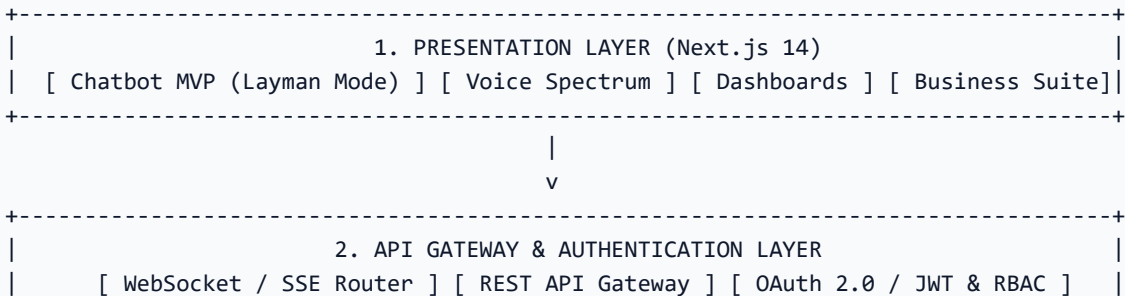
1. Executive Summary & Vision Statement

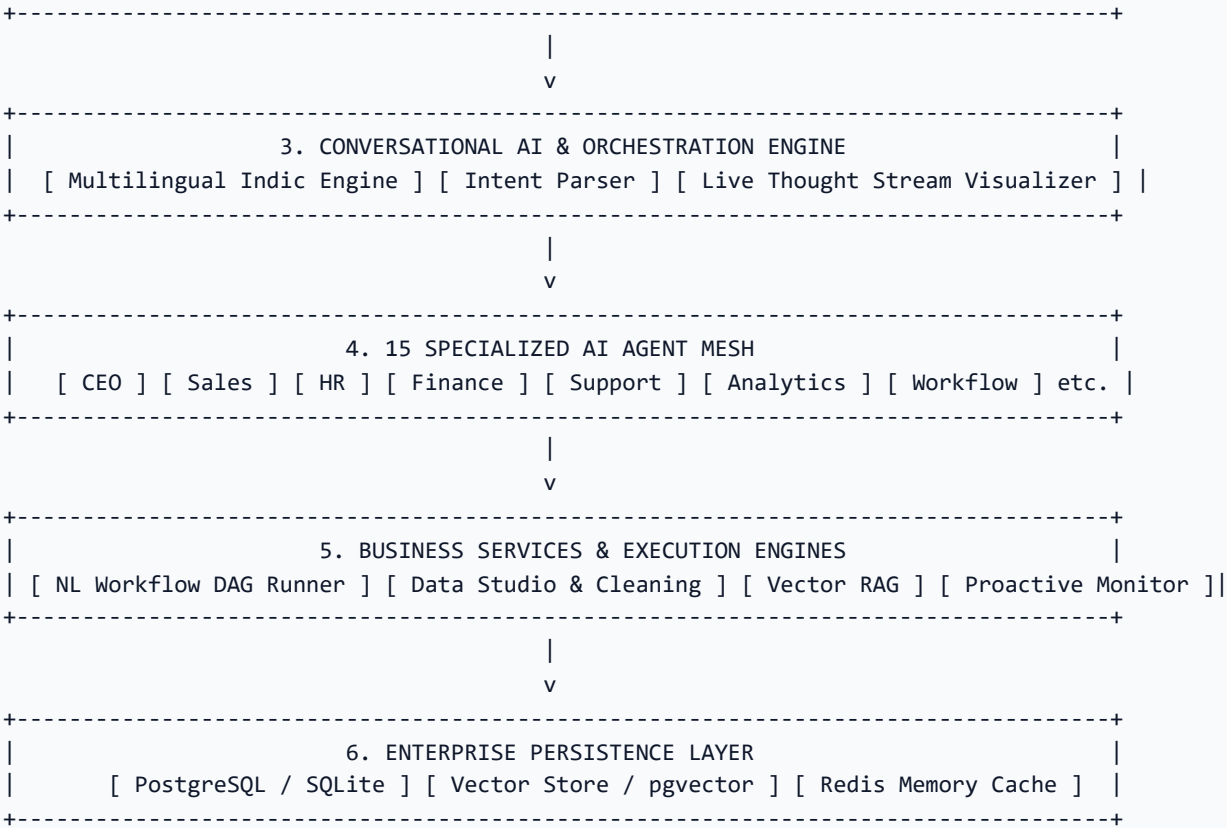
FlowMind AI is an enterprise-grade Multilingual AI Business Operating System. It unifies business operations (CRM, HR, Finance, Inventory, Support, Workflow Engine, Data Studio, RAG Base, Proactive Monitoring) behind an intuitive, layman-accessible **AI Chatbot MVP**.

2. SDLC Architecture & Quality Gates

SDLC Phase	Key Deliverable	Quality Gate & Acceptance Criteria
Phase 1: Requirement Analysis	Layman User Stories & Multilingual Taxonomy	UX Accessibility & Persona Signoff
Phase 2: System Architecture	Multi-Agent Protocol, DB Schemas, DAG Planner	Response Latency < 500ms
Phase 3: Core Chatbot MVP	Chat UI, Indic Engine, Thought Stream, Action Cards	Unit Coverage > 85%, Layman Clarity Test
Phase 4: Business Suite	CRM, HR, Finance, Data Studio, RAG Hub	End-to-End Execution Verification
Phase 5: Security & RBAC	Multi-tenancy, Sanitization, Audit Logs	Penetration Test & OWASP Compliance
Phase 6: Deployment & Ops	Docker Packaging, CI/CD, Telemetry	99.9% Uptime SLA Target

3. High-Level Architecture Block Diagram





4. Layman Chatbot MVP Interaction Architecture

The Chatbot MVP is specifically designed so non-technical users can interact, observe agent collaboration, and approve actions with complete transparency.

1. Natural Language or Voice Input

User speaks or types in native language (e.g. "రేపు ఉదయం 10 గంటలకు నేల్స్ మీటింగ్ పెట్టు" or "Send welcome email to new leads & remind me in 2 days").

2. Multilingual Intent & Speech Router

Auto-detects language script (Telugu, Hindi, Tamil, Kannada, Malayalam, English) and maps user request to canonical business intent.

3. Live "Thought Stream" Visualizer

Renders real-time progress pills showing agent collaboration: [Sales Agent: Found 3 Leads] → [Email Agent: Drafted Welcome Letter] → [Scheduler: Set Reminder].

4. Interactive Action Confirmation Card

Presents a clean card requesting explicit confirmation before taking action: [Approve & Execute], [Modify], [Cancel].

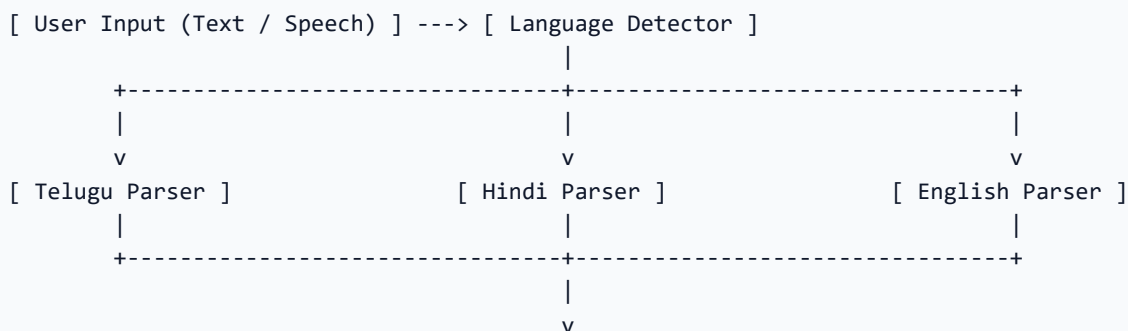
5. Execution & Natural Response

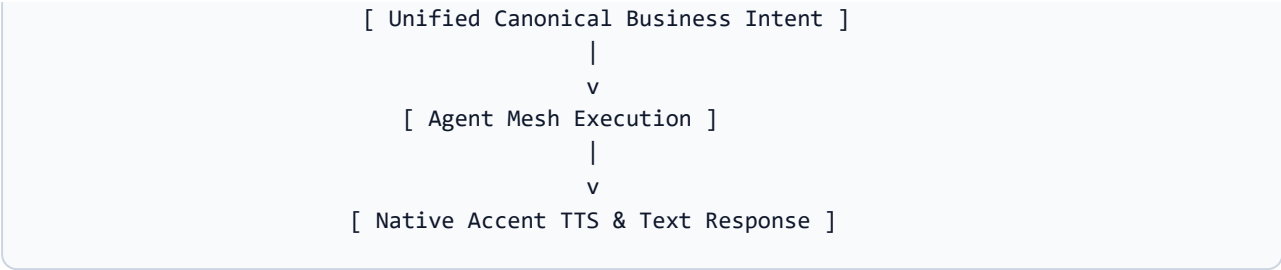
Executes workflow step-by-step and responds with natural voice/text summary in user's native language.

5. Multilingual Engine Architecture

Native script processing for 6 core languages with expandable translation dictionary:

- **English** (en) - Global Standard
- **Hindi** (hi - हिन्दी) - North / Central India
- **Telugu** (te - తెలుగు) - Andhra Pradesh / Telangana
- **Tamil** (ta - தமிழ்) - Tamil Nadu
- **Kannada** (kn - ಕನ್ನಡ) - Karnataka
- **Malayalam** (ml - മലയാളം) - Kerala



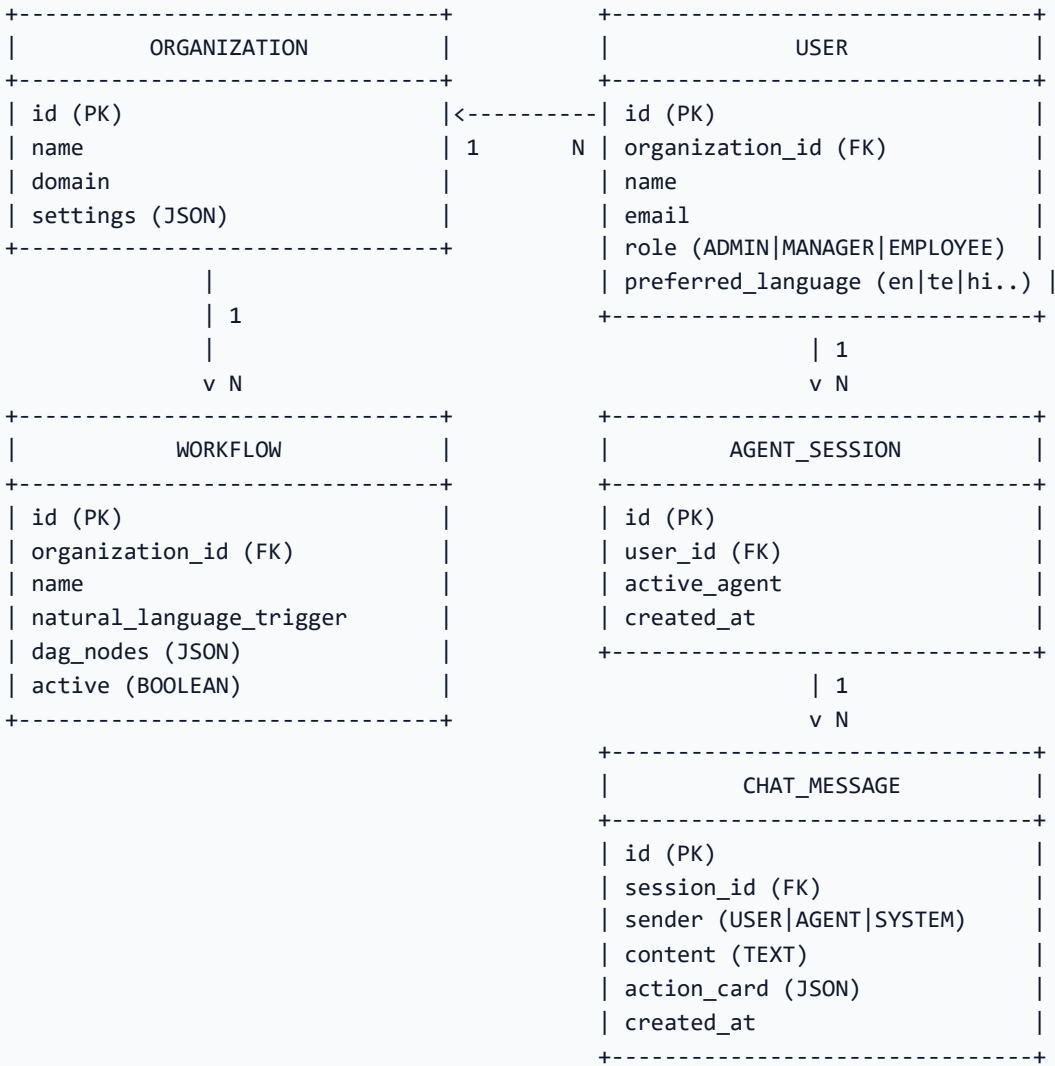


6. 15-Specialized AI Agent Mesh

Specialized agents execute domain-specific tasks rather than relying on a single monolith model:

Agent Name	Domain Ownership	Key Tools & Capabilities
CEO Agent	Executive Governance	Strategic health summaries, company alignment, cross-agent oversight
Sales Agent	CRM & Lead Pipeline	Lead scoring, deal stage tracking, follow-up scheduling
Marketing Agent	Campaigns & Copywriting	Ad copy generator, campaign performance, email marketing
Finance Agent	Invoices & Payroll	Invoice generation, overdue payment alerts, tax calculation
HR Agent	Employee Management	Leave approval, payroll processing, employee onboarding
Support Agent	Customer Helpdesk	Ticket routing, SLA monitoring, resolution generator
Workflow Planner	Automation Engine	Natural language to DAG workflow parser, conditional builder
Document RAG Agent	Knowledge Base	Document chunking, vector similarity search, citation linking

7. Database Entity-Relationship Model (ERD)



8. Security, RBAC & Enterprise Compliance

- Multi-Tenant Data Isolation:** Tenant ID scoped on every database query and cache key.
- Role-Based Access Control (RBAC):** Three-tier permissions: ADMIN (full access), MANAGER (departmental access), EMPLOYEE (self-service & limited chat).
- Prompt Injection Protection:** Dual-layer input sanitizer strips unauthorized system instructions and escapes raw user queries.
- Audit Logging:** All workflow executions, agent decisions, and financial operations are written to immutable audit logs.